

Basic Principles of Medicine 1
Module: Foundations to Medicine
DLA

Amino Acids: Structure and Function

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DLA: Amino Acids - Structure and Function

Objectives

- Before lecture ‘Structure and function of proteins’

SOM.MK.I.BPM1.1.FTM.3.BCHM.0078	Classify amino acids based on their R groups (with examples for each group).
SOM.MK.I.BPM1.1.FTM.3.BCHM.0079	Name the amino acids with a charged side chain, the amino acids containing a hydroxyl group and the amino acids containing sulphur.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0081	Discuss the role of histidine residues in buffering in hemoglobin.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0083	Discuss the importance of derivatives of amino acids (GABA, nitric oxide, histamine, serotonin and catecholamines).

Recommended Reading

- **Lippincott Illustrated Reviews, Biochemistry, 8th Edition:**

<http://sgu.idm.oclc.org/login?url=https://meded.lwwhealthlibrary.com/book.aspx?bookid=3073>

- Chapter 1: Amino Acids and The Role of pH (pages 1 – 13)

- Concept map: page 11
- Questions: 1.1 – 1.3

[Amino Acids and the Role of pH | Lippincott® Illustrated Reviews: Biochemistry, 8e | Medical Education | Health Library \(lwwhealthlibrary.com\)](#)

- Chapter 2: Protein Structure (page 15)
- Chapter 21: Amino Acids: Conversion to Specialized Products (Pages 317-319)

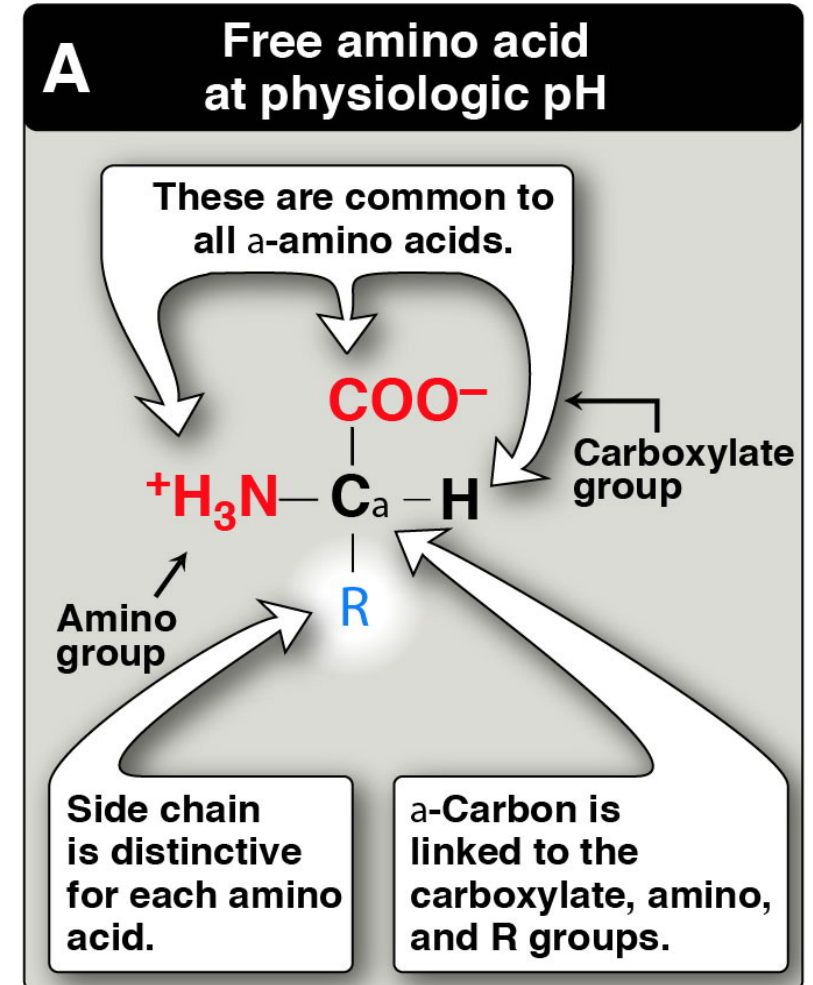
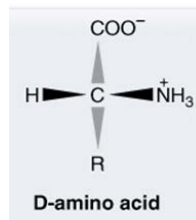
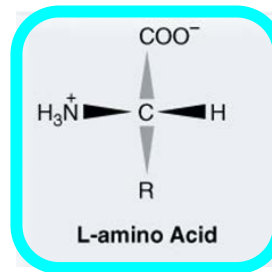
- **Marks' Basic Medical Biochemistry:**

- Chapter 6:

[Amino Acids in Proteins | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Premium Basic Sciences | Health Library](#)

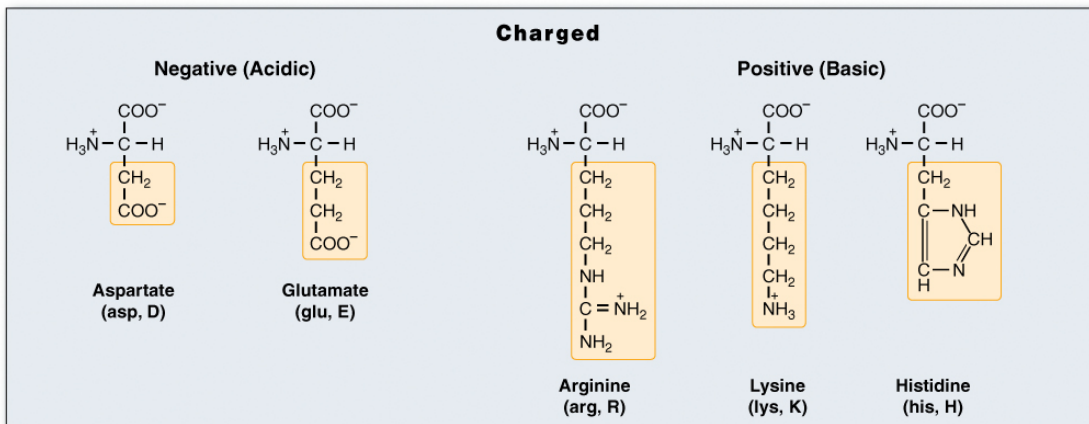
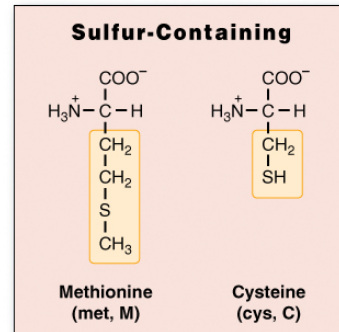
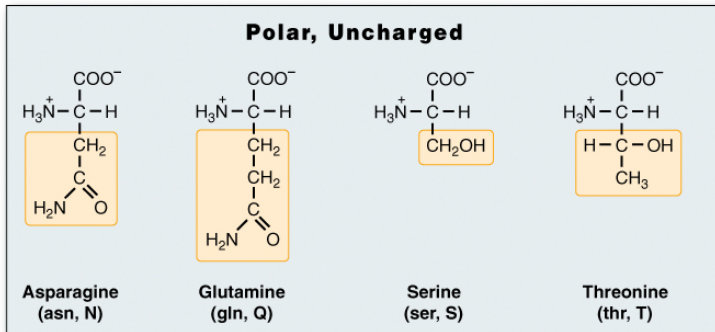
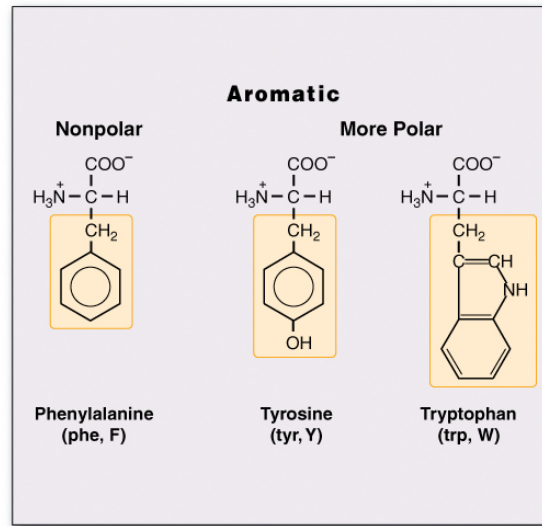
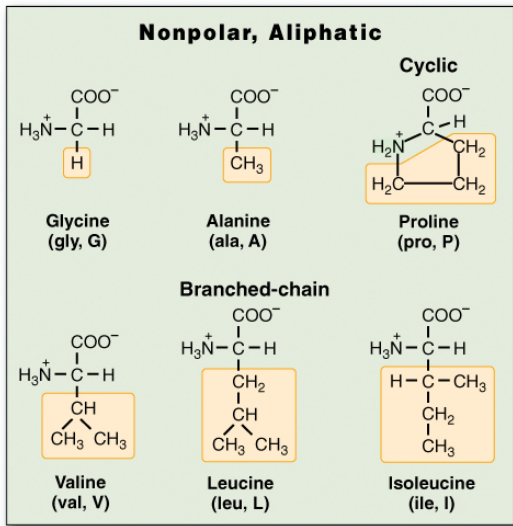
General Structure of Amino Acids

- An α - carbon atom: Asymmetric (except glycine) – linked to:
 - A carboxylic acid group
 - An amino group
 - An H atom
 - A variable R group (side chain): any molecule can replace the 'R'. For instance, glycine has an -H instead and alanine (-CH₃)
- 20 standard amino acids in proteins
 - Standard amino acids found in proteins (contain genetic codon)
- Proteins contain **L- amino acids**
- D-amino acids found in some antibiotics & in bacterial cell wall.



AAs are classified based on their side chains (R-group).

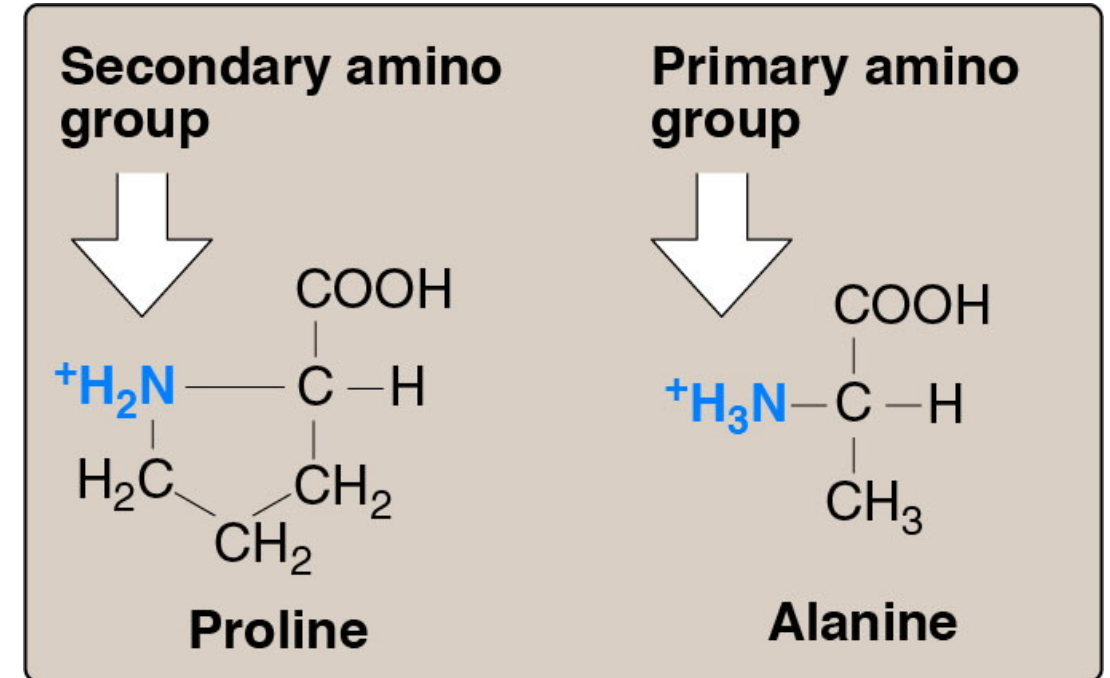
Classification of Amino Acids



- Grouped by polarity and structural features of side chains.
- The side chains are highlighted. The single and three-letter codes are indicated
- Aromatic amino acids:** Trp and Tyr absorb UV light at 280 nm wavelength.
- Methionine (**sulfur containing**) is also nonpolar
- Cysteine (**sulfur containing**) is also uncharged polar side chains at neutral pH.
- Asparagine: Derivative of aspartate
- Glutamine: Derivative of glutamate
- Serine, tyrosine & threonine** residues contain a **hydroxyl group** - which may be phosphorylated or glycosylated.
- Hydrophobic AAs:** Valine, leucine, isoleucine, phenylalanine, tryptophan and to some extent tyrosine.

Proline: α -helix breaker

- Secondary amino group and rigid side chain
- Peptide bonds with proline take a specific shape
 - Leads to bends in protein chain
 - Interrupts α -helical structure
 - Proline is an α -helix breaker
 - **Collagen** is rich in proline residues

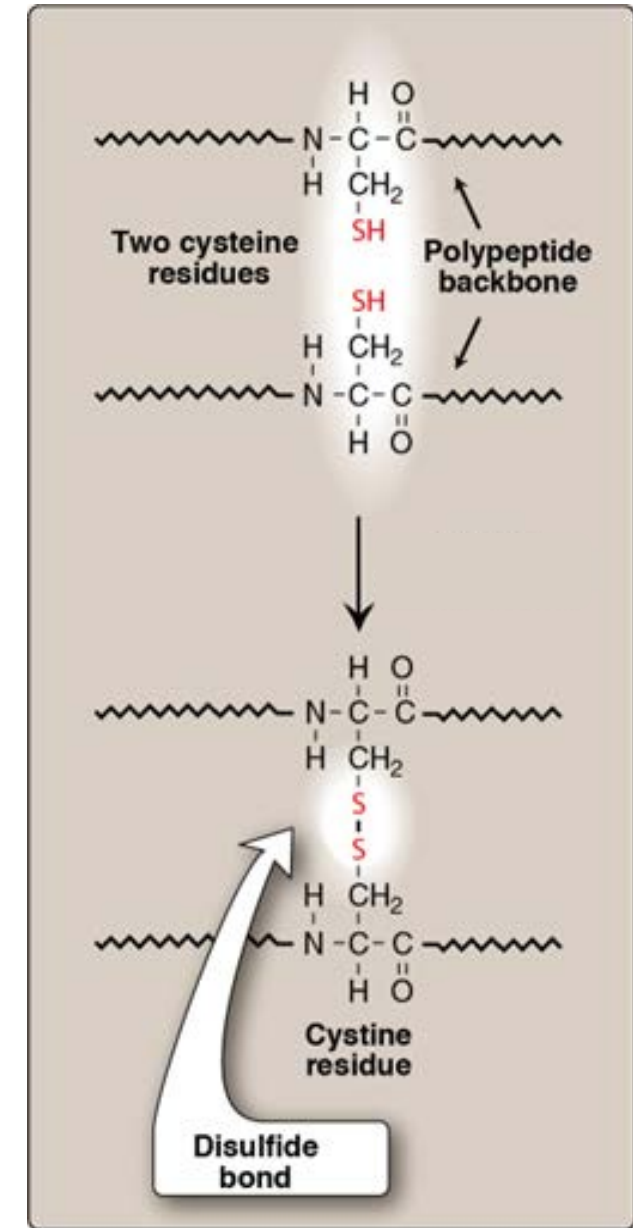


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Secondary amino group of proline, compared to primary amino group of other AAs.

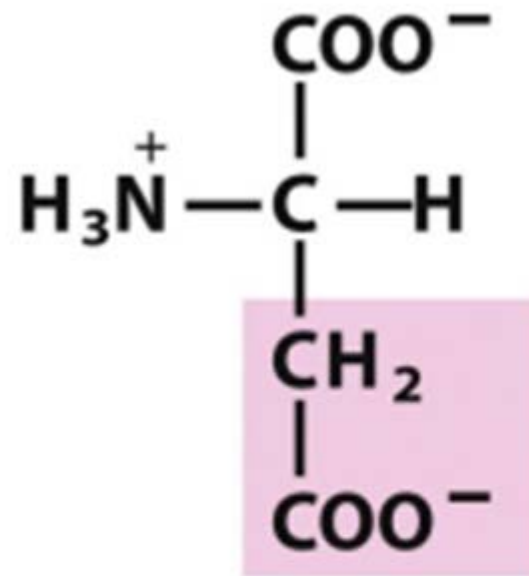
Cysteine: Sulfhydryl (SH) group

- Two cysteine residues can be used to form a **disulfide bond** in proteins.
 - Disulfide bonds are involved in stabilizing many proteins
- The thiol (-SH) group also plays an important role in “active site” of enzymes

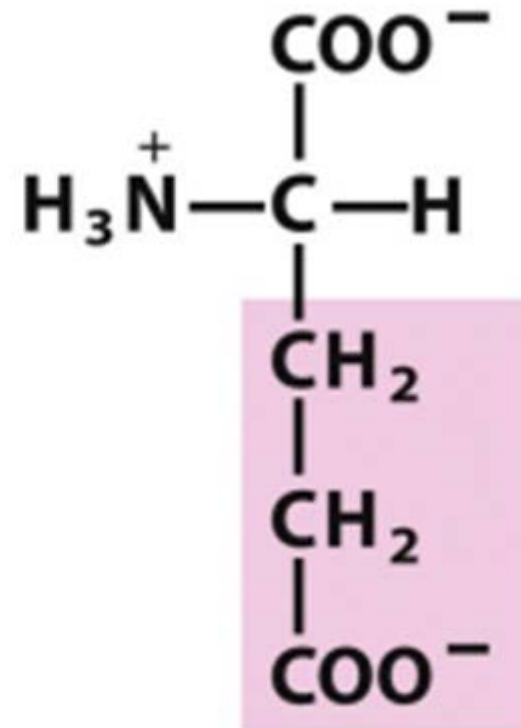


Aspartate and Glutamate:

Negatively (acidic) charged at physiological pH



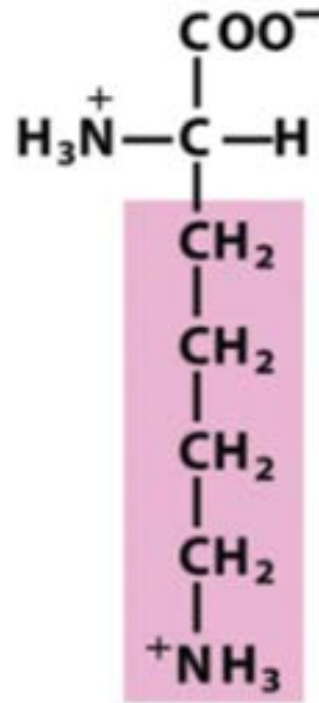
Aspartate



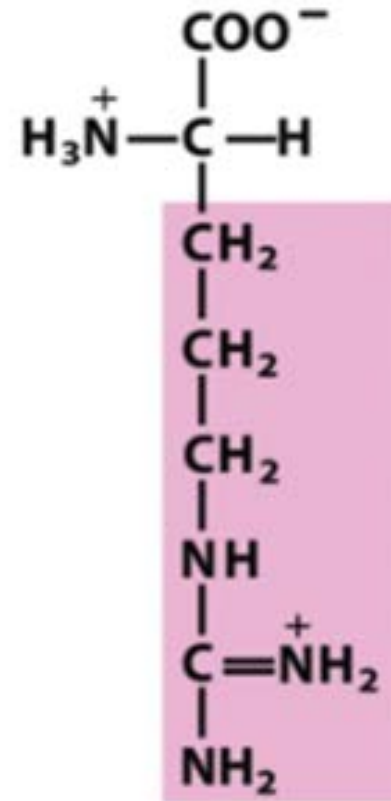
Glutamate

Arginine and Lysine: Positively (basic) charged (Sometimes histidine - dependent on exact pH)

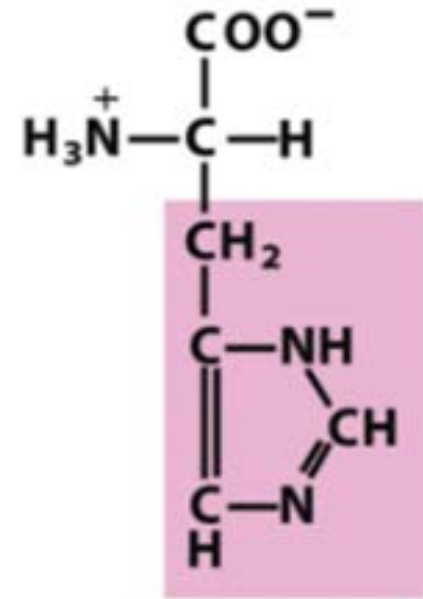
- Side chains of basic amino acids accept protons.
- Lysine and arginine are positively charged at physiologic pH
- Abundant in histones which bind negatively charged DNA



Lysine



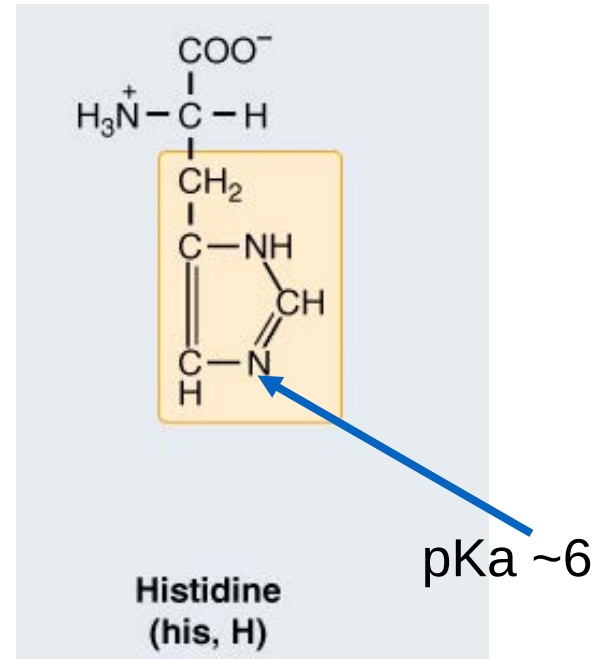
Arginine



Histidine

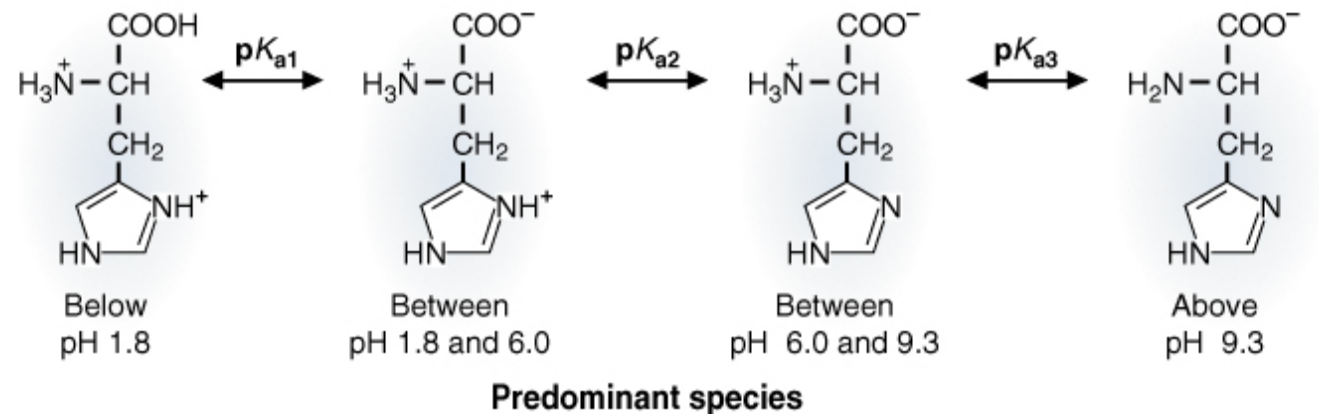
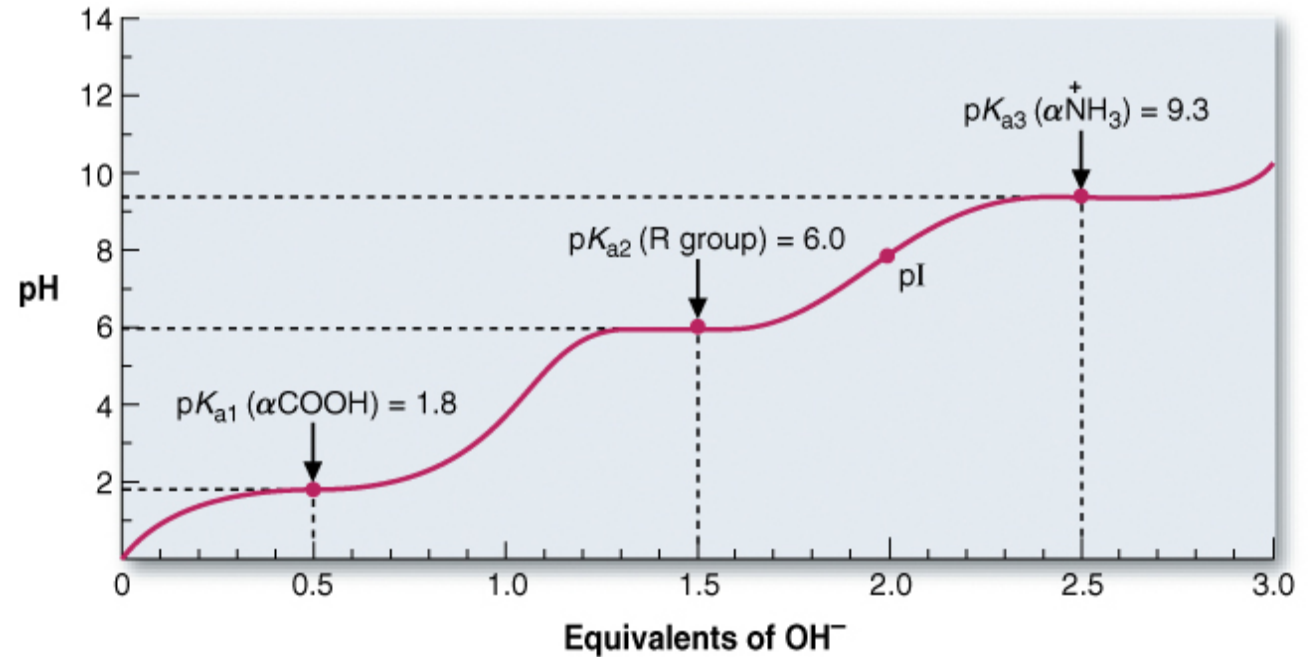
Histidine: Role as Buffer

- Buffer resists a change in pH following addition of an acid or base.
 - Contains a **weak acid** and its **conjugate base**
 - Greatest buffering capacity in the pH range near its **pK_a**
- Histidine in proteins can be positively charged (protonated) or neutral, (depends on ionic environment in protein)
- Contributes to **buffering role** in proteins like hemoglobin.
- Histidine side chain can donate/accept protons in the physiologic pH range (7.35 to 7.45) - Functions as a buffer
- Most amino acids are not suitable as physiological buffers except histidine.



Titration curve of Histidine: pKa2 defines the buffer region

- Histidine has imidazole ring (pKa 6) which allows buffering
- The pKa of histidine in proteins is about 6-7 due to chemical environment of the protein
- **Histidine** is favorable to function as a **buffer**



Functions of Amino acids

Peptides and proteins

Purine and pyrimidine nucleotides

Peptide hormones

Melanin

**Heme, creatine, carnitine,
sphingolipids**

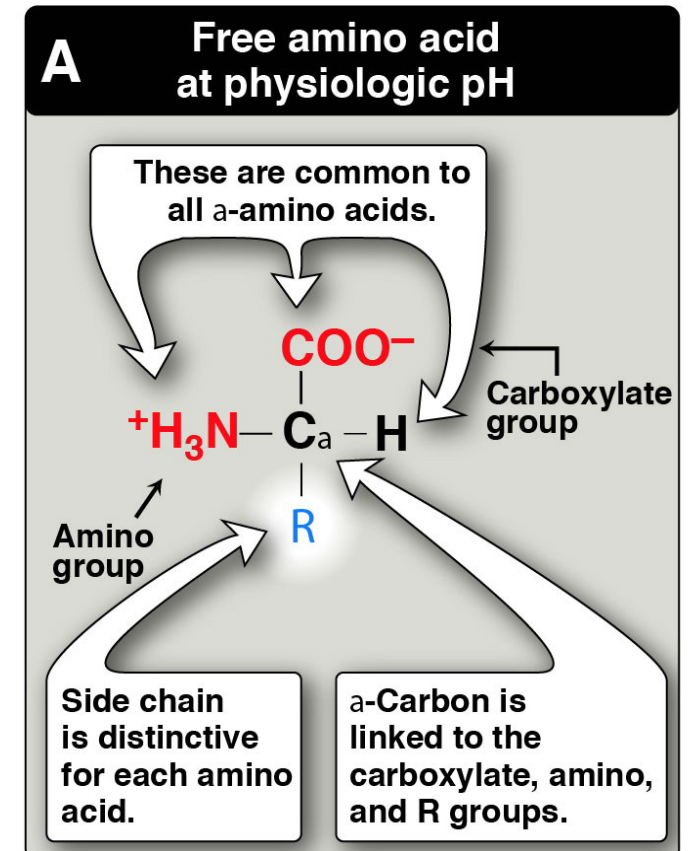
Urea (waste product)

**Physiologically active amines:
Catecholamines;
Neurotransmitters**

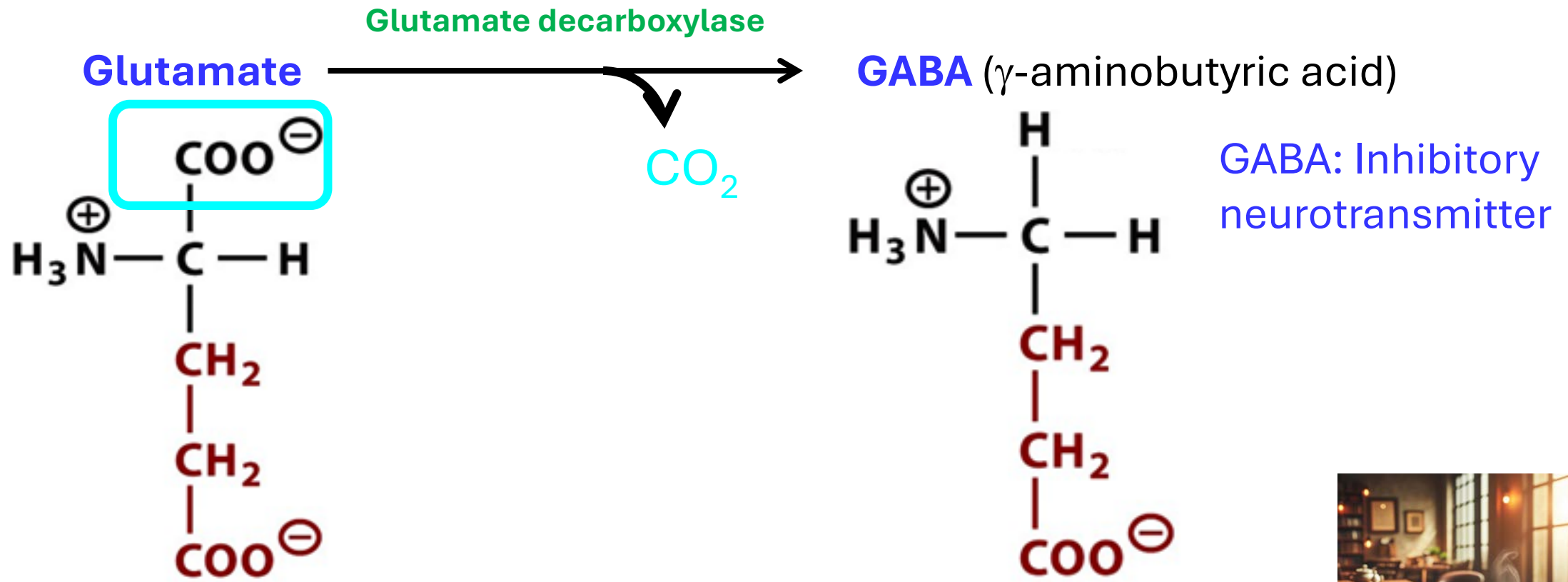
Nitric oxide

Formation of Biologically Active Amines

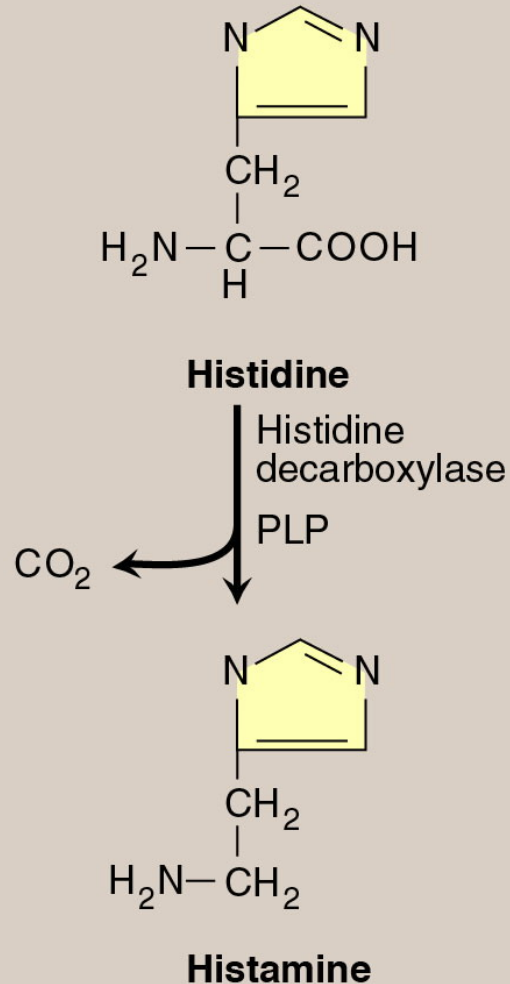
- α -amino acids are precursors for biologically active amines (biogenic amines).
- Amino acids or their derivatives act as small molecule neurotransmitters, or hormones.
- **Amino acid decarboxylase: α -carboxyl group removed as CO_2 (decarboxylated); Amino group retained (amines)**



GABA (γ -aminobutyric acid): from Glutamate



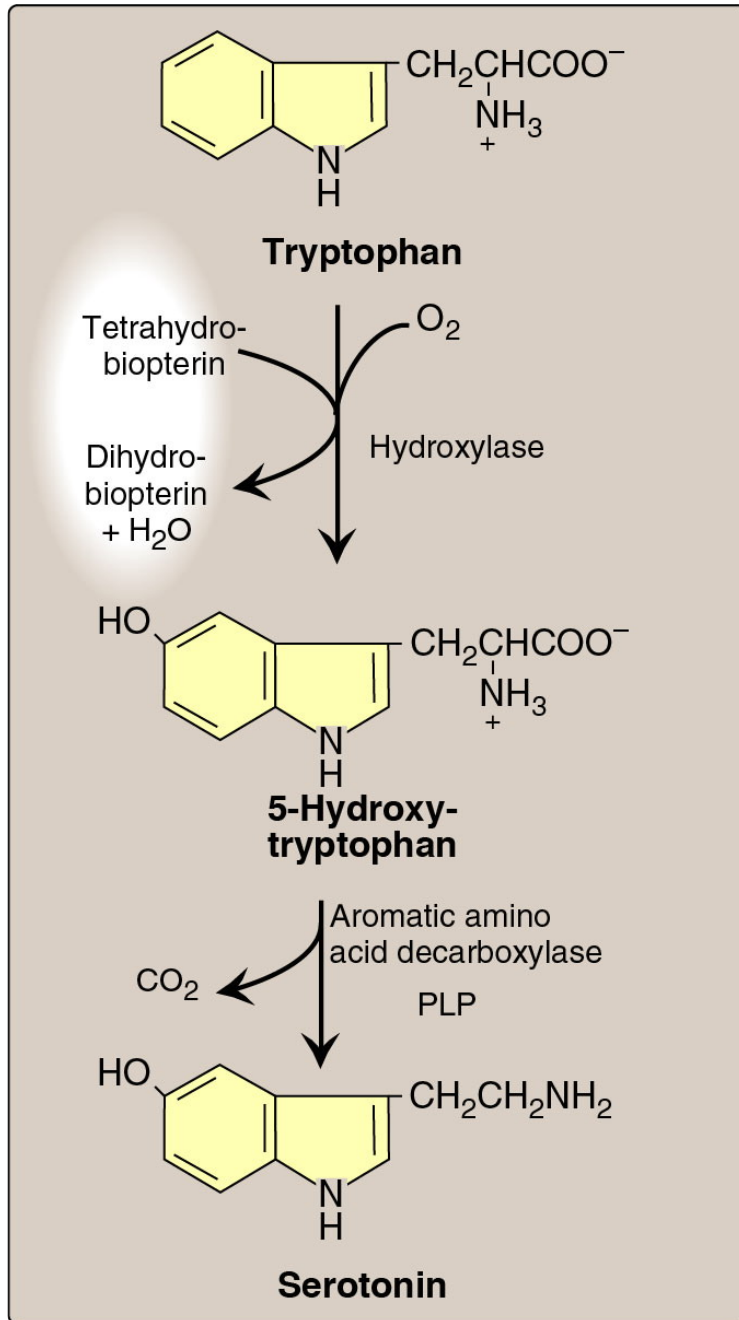
Histamine: formed by decarboxylation of histidine



- Histamine functions:
 - Mediates allergic and inflammatory responses
 - Stimulates gastric secretion
 - Neurotransmitter



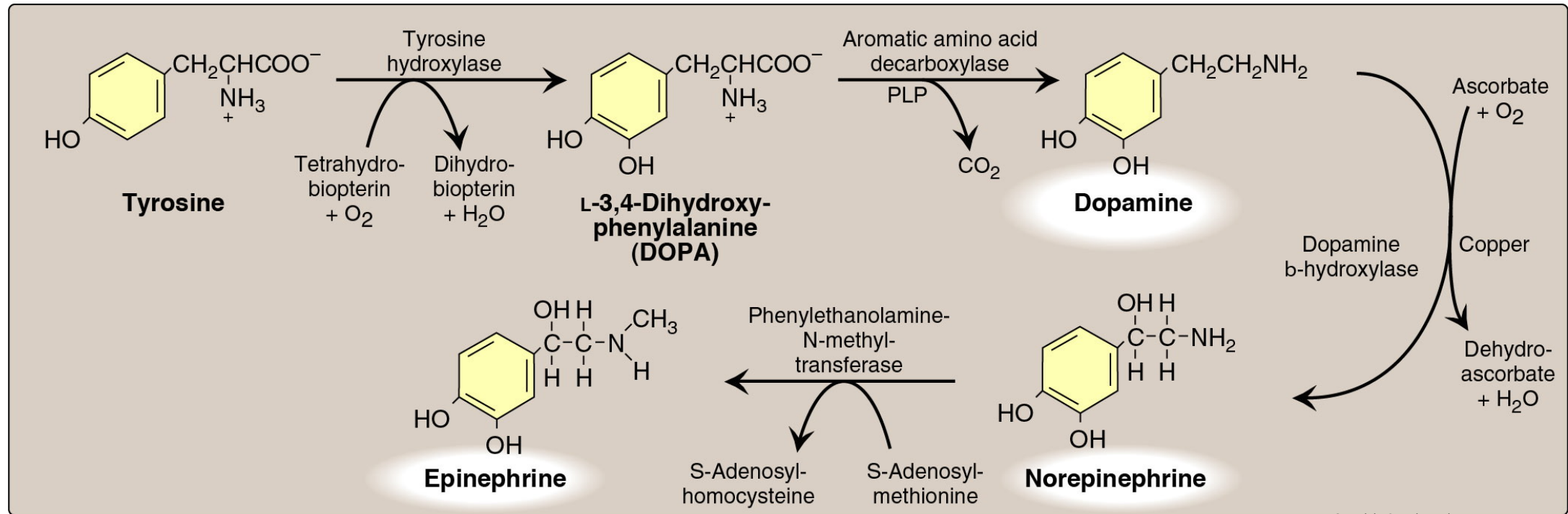
Serotonin: from Tryptophan



Serotonin physiological roles:

- Pain perception
- Affective disorders
- Regulation of sleep, temperature and blood pressure
- Cognitive functions, and mood (causes a feeling of well-being)

Catecholamines: Dopamine, norepinephrine and epinephrine (or adrenaline) formed from Tyrosine

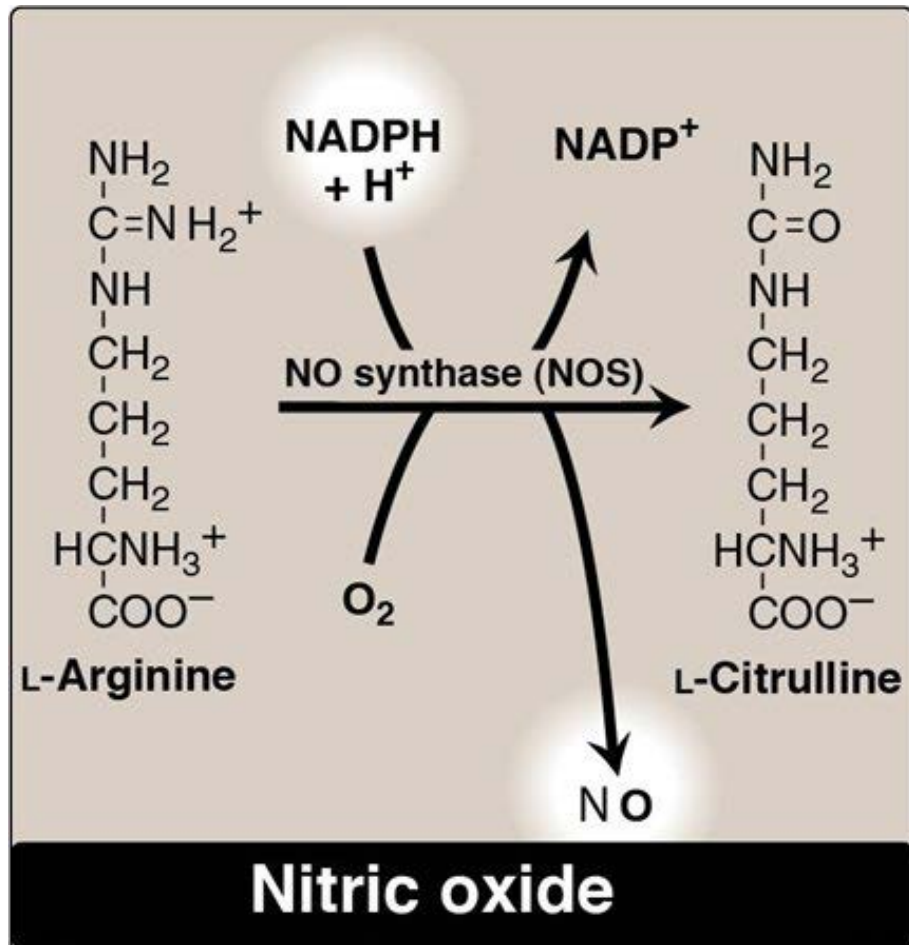


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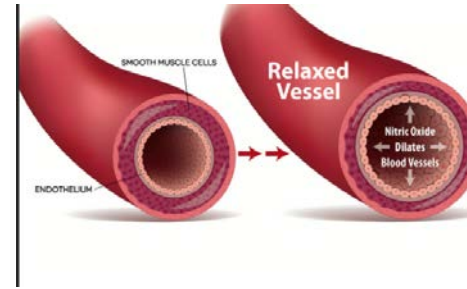
- **Dopamine and norepinephrine: Neurotransmitters** in brain and autonomic nervous system.
- **Epinephrine and norepinephrine: Hormones** from **adrenal medulla**
- **Parkinsons disease** is associated with loss of formation of dopamine in the brain.
 - L-DOPA prescribed in Parkinson disease to produce dopamine



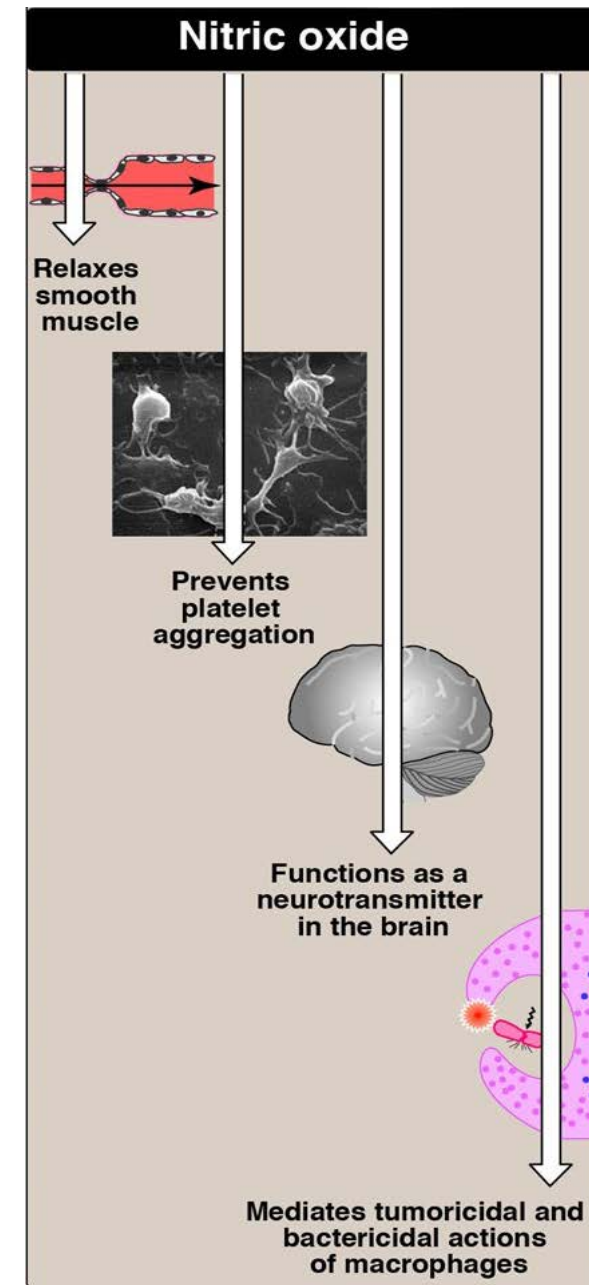
Nitric oxide: from Arginine



Synthesis of Nitric Oxide (NO)



- From **arginine**
- Functions:
 - Vasodilator
 - Prevents platelet aggregation
 - Neurotransmitter
 - Forms reactive nitrogen species during phagocytosis



Functions of Nitric Oxide (NO)

Summary

- Classification of amino acids
- Specific roles of side chains in amino acids
 - DO NOT MEMORIZE structures
- Specialized products from amino acids:
 - GABA
 - Histamine
 - Catecholamines
 - Serotonin
 - Nitric oxide

Thank you.

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